Bubbles and Priors: A Language-Model Experiment

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Abstract. We replicate the finite-horizon experimental asset market of Dufwenberg et al. (2005) inside a deterministic browser-based simulator and extend it with the expected-utility messaging population of Lopez-Lira (2025). We then introduce a new paradigm—*Wang 2026*—in which each Lopez-Lira utility agent's starting subjective valuation is drawn, once per run, from a live ChatGPT call whose prompt encodes the agent's risk preference, bias mode, and the presence of experienced counterparties. The returned psychological anchor $\hat{V}_i$ is clamped to $[0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]$, seeds the agent's prior on its first decision tick, and then yields to the standard Lopez-Lira belief-blending channel. The research question is whether the asset ultimately ends up in the hands of the agent with the highest anchored valuation. The testbed reports per-tick matching, a normalised valuation gap, Haessel R^2, normalised deviation, amplitude, turnover and allocative efficiency; the merged Experienced-plus-Utility population is split as $\lceil (N - N_{\text{fixed}})/2 \rceil$ experienced and the remainder utility. The simulator preserves reproducibility through a seeded mulberry32 PRNG and a deterministic fallback when the language-model call fails, so every finding can be replayed without network access. This paper documents the replication, the new anchor channel, and the market-quality instruments used to answer the allocation question.

Keywords: experimental asset markets, bubbles, expected utility, large language models, ChatGPT, allocative efficiency.

JEL Codes: C92, D84, G12, G41.

I. Introduction

Experimental asset markets in the tradition of Smith et al. (1988) have produced one of the most replicated departures from rational-expectations pricing in all of economics: even when subjects know, in closed form, that the fundamental value of a finite-lived asset declines deterministically with the remaining dividend stream, transaction prices systematically overshoot the fundamental before collapsing near the terminal period. Dufwenberg et al. (2005)—hereafter DLM 2005—established a by now canonical result in this literature: when two-thirds or more of the subject pool consists of traders who have played the game before, the bubble disappears. Experience, rather than cognition or incentives, is what eventually kills the bubble.

A parallel and much more recent literature replaces the human subjects with large language models. Lopez-Lira (2025) proposes an expected-utility framework in which each LLM-flavoured agent carries a persistent valuation bias b_i, a risk preference over wealth $U \in \{U_L, U_N, U_A\}$, and a belief-blending channel that lets agents exchange private valuations and update a trust score for each sender by exponential moving average. Lopez-Lira's agents optimise expected utility over a small action set—hold, cross the bid, cross the ask, post passively—and the resulting population can be dialled between skeptical and naive, truthful and deceptive, risk-averse and risk-loving by simple categorical knobs. The Lopez-Lira design is a natural vehicle for studying how a heterogeneous LLM population prices a DLM asset, but it leaves open a psychological question the original DLM protocol never asked: if one replaces the deterministic prior $\mathrm{FV}_t \cdot (1 + b_i + \varepsilon)$ with

*Computational replication and source code are available at `https://github.com/Mon-ius/bubble`. The author thanks the maintainers of the original DLM and Lopez-Lira testbeds for making their designs easy to re-implement. All errors are the author's own.

a private subjective valuation drawn from a language model that is told who the agent is and what the market looks like, does the asset then migrate, over the course of the run, into the hands of the agent with the highest such valuation?

This paper contributes a computational testbed in which the two designs share one continuous double-auction market, and a new paradigm—which we label *Wang 2026*—that seeds each utility agent's prior with exactly one ChatGPT call at run start. Concretely, when the user selects Wang 2026 and supplies an API key, the simulator fires one chat completion per utility agent in parallel, using a prompt that encodes the agent's risk preference, bias mode, and the presence of DLM experienced counterparties in the same market. The returned number, clamped to $[0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]$, is written to the agent's `psychAnchor` slot and consumed on the first decision tick in place of the deterministic prior; subsequent ticks revert to the Lopez-Lira prior plus belief blending. In the notation of the replicated model,

$$\hat{V}_i \;\leftarrow\; \mathrm{ChatGPT}(\mathrm{agent}_i), \qquad \mathrm{prior}_{t=0}^{(i)} \;=\; \hat{V}_i \cdot (1+\varepsilon_i), \qquad \hat{V}_i \in [0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]. \tag{1}$$

The deterministic fallback—used when no key is supplied or the network call fails—drops equation (1) and resumes the Lopez-Lira prior $\mathrm{FV}_t \cdot (1 + b_i + \varepsilon)$ unchanged. Because the run then proceeds through a deterministic seeded PRNG, every simulation can be replayed identically with or without the anchor channel.

The research question is easy to instrument. Let $i^\star = \arg\max_{i \in U} \hat{V}_i$ be the utility agent with the highest anchored valuation and $j^\star(t) = \arg\max_i \mathrm{inventory}_i(t)$ the top holder at tick t; a *match* at t is $\mathbb{1}\{j^\star(t) = i^\star\}$ and the *normalised gap* is $g(t) = (\hat{V}_{i^\star} - \hat{V}_{j^\star(t)})/\hat{V}_{i^\star}$. Both quantities are reported every tick alongside Haessel R^2, normalised deviation, amplitude, turnover, and allocative efficiency, so classical bubble statistics stay visible beside the new Wang metric block.

Two features distinguish the Wang 2026 paradigm from a naïve "LLM-as-trader" pipeline. First, the language model is consulted exactly once at run start; after the anchor is absorbed, the agent is again a pure expected-utility maximiser and the mulberry32 RNG alone controls the downstream dynamics. Second, the anchor is a *valuation* and not an order: the agent's quoting rule, fill probability, and utility function are unchanged, and only its starting prior is moved. These two choices isolate the specific question "does psychology predict ownership" from the adjacent question "does an LLM trade well."

Section II places the simulator in the literature on experimental asset markets and LLM-based agents. Section III documents the market structure and the sampling stage. Section IV gives the precise decision rule for each agent type. Section V presents the Wang 2026 extension: the anchor call, the clamp, the consumption path, and the research-question instrumentation. Section VI reports preliminary findings, and Section VII concludes.

II. Related Literature

The simulator sits at the intersection of four strands of literature: the experimental-bubble tradition, the heterogeneous agent-based finance tradition, the expected-utility decision-theoretic tradition, and the emerging work on large language models as economic agents.

Experimental bubbles.. The taxonomy of the market in this paper is directly inherited from Smith et al. (1988), who introduced the declining-fundamental, finite-horizon asset and documented the characteristic bubble-and-crash trajectory in a continuous double auction. Dufwenberg et al. (2005) show that the Smith–Suchanek–Williams bubble is fragile to *experience* rather than to instructions, incentives, or trading institutions: once two-thirds of the subjects have played the game before, transaction prices track the fundamental staircase from the first period onward. This result motivates the horizon-discount rule used by the simulator's *Experienced* (E) agent, in which an experience level $\varphi(h)$ ramps belief from FV_t in early rounds to zero in the last three periods.

Heterogeneous agent-based finance.. Three classical rules supply the pinned background of the simulator: a *Fundamentalist* (F) that mean-reverts toward FV_t in the Smith–Suchanek–Williams lineage; a *Trend follower* (T) that implements the positive-feedback strategy of De Long et al. (1990) by extrapolating a six-tick slope; and a *Random* (R) zero-intelligence trader after Gode and Sunder (1993) that posts uniform bids on $[0.7, 1.1]\,\mathrm{FV}_t$ and uniform asks on $[0.9, 1.3]\,\mathrm{FV}_t$. Together these three rules provide a rational anchor, a mo-

mentum amplifier, and a liquidity floor; the DLM experienced agent then contributes the horizon-conditional bubble-suppression channel.

Expected utility and LLM traders.. Lopez-Lira (2025) introduces a utility-maximising LLM agent whose prior is a perturbation of FV_t, whose action space is {hold, cross bid, cross ask, passive bid, passive ask}, and whose choice rule picks the action with the highest expected utility, $\mathrm{EU}(a) = p_{\text{fill}} \cdot U(w_1) + (1 - p_{\text{fill}}) \cdot U(w_0)$. Agents carry one of three risk-preference functionals $\{U_L(w), U_N(w), U_A(w)\} = \{(w/w_0)^2,\ w/w_0,\ \sqrt{w/w_0}\}$ and exchange private valuations through a lightweight messaging bus. A pairwise trust score $\tau_{r \to s} \in [0, 1]$ is updated by exponential moving average whenever a claimed valuation is observed against the period's volume-weighted average trade price. The present paper inherits this framework without modification and reimplements it exactly as described, so that any difference from a pure Lopez-Lira run is attributable to the ChatGPT-anchored prior rather than to a change in the decision rule.

Language models as economic subjects.. A small but rapidly growing literature studies whether large language models reproduce well-documented behavioural regularities when embedded in experimental decision tasks. The Wang 2026 paradigm is a deliberately narrow use of this approach. It does not ask the language model to trade; it does not expose the order book or the history of transaction prices; it does not grant the model the authority to choose an action from the action set. It asks one question—"what is your private subjective valuation per share right now, in cents?"—with a prompt that names the agent's risk preference, bias mode, and the composition of the market. The returned number is then clamped and treated as a prior in the downstream expected-utility computation. This design isolates the psychological channel ("how would the model price the asset, in light of who it is?") from the trading channel ("what orders would the model place?"), and in doing so connects the experimental bubble literature to the psychology-of-valuation literature via a single numeric quantity.

Market-quality instruments.. The testbed reports four standard bubble statistics—Haessel's R^2 (Haessel, 1978), normalised absolute deviation, amplitude, and turnover, following SSW and Stöckl et al. (2010)—alongside allocative efficiency $\mathrm{AE} = (\sum_i \hat{V}_i \cdot \text{inventory}_i)/(\max_i \hat{V}_i \cdot Q)$, the quantitative form of the Wang 2026 research question: when $\mathrm{AE} = 1$ the asset has migrated to the agent with the highest anchored valuation.

III. Experimental Design

A. The asset and the fundamental

The asset is a finite-lived claim on an iid per-period dividend that pays either zero or twenty experimental cents with equal probability. Letting d denote the period-t dividend draw,

$$d \sim \text{Uniform}\{0, 20\}, \qquad \mathbb{E}[d] = 10, \qquad T = 10. \tag{2}$$

The asset lives for $T = 10$ periods and pays nothing thereafter, so its risk-neutral fundamental value at the start of period t is

$$\mathrm{FV}_t \;=\; \mathbb{E}[d] \cdot (T - t + 1) \;=\; 10 \cdot (T - t + 1), \tag{3}$$

that is, a linearly declining staircase from $\mathrm{FV}_1 = 100$ to $\mathrm{FV}_T = 10$ with a terminal value of zero after the final dividend pays. Equation (2) and equation (3) reproduce the paper constants of Dufwenberg et al. (2005) exactly; both the number of periods and the dividend mean are surfaced read-only in the simulator's "Paper constants" panel and are not user tunable, so that every comparison across populations holds the market structure fixed.

B. Continuous double auction with period discretisation

The simulator discretises each period into $K = 18$ decision rounds, so a run comprises $T \cdot K = 180$ ticks. Every tick, each agent posts at most one order, routed to a price–time-priority order book; a crossing pair matches at the resting price. At the end of each period, outstanding quotes are withdrawn and the dividend is drawn from (2) and credited to each holder. The $K = 18$ discretisation is pinned in the "Simulator constants" panel and is not user tunable.

C. Sampling and endowments

At $t = 0$ the simulator draws a flat list of per-agent specifications from the user's chosen population mix. For each slot, cash is sampled uniformly from $[800, 1200]$ experimental cents and initial inventory is drawn uniformly from $\{2, 3, 4\}$ shares, so that total endowments are heterogeneous but bounded. Every slot is also assigned a randomly drawn personal name (from an internal name list) and a type label such as F1, U3, E2, which is used in the agents panel and the trust matrix for human legibility. Crucially, the sampling RNG is derived from seed $\oplus$ `0xA5A5A5A5` and is independent of the engine RNG; this is the invariant that lets a user edit cash or inventory on an existing draw and re-run without re-sampling agent types, while still producing the same per-tick trading sequence as a matched run without the edit.

D. Populations

Four canonical population presets are available. Table 1 summarises their composition and the qualitative outcome each is intended to showcase.

Table 1: Population presets

Preset	Composition	Expected outcome
Inexperienced	2 T + 2 R + 1 F + 1 E	Classic bubble and crash
Experienced	3 E + 2 F + 1 T	Tight convergence to FV_t
Mixed	2 F + 2 E + 1 T + 1 R	Closest tracking of FV_t
Utility	6 U	Lopez-Lira default

The "Number of Agents" slider rescales the E and U slots proportionally while pinning the fixed F/T/R background at $N_{\text{fixed}} = 4$. The paradigm switch in the navbar overrides these presets: clicking DLM 2005 fills the user-controllable slots with experienced agents ($E = N - N_{\text{fixed}}$, $U = 0$); clicking Lopez-Lira 2025 fills them with utility agents ($E = 0$, $U = N - N_{\text{fixed}}$); clicking Wang 2026 splits the slots as $\lceil (N - N_{\text{fixed}})/2 \rceil \to E$ and the remainder to U, so that both subject types share one market and the extended panels stay populated. The rounding is asymmetric on purpose: at capacity $c = 1, 3, 5$ the E block wins the odd slot, which preserves the bubble-suppressing channel in small populations where it matters most.

E. Reproducibility

Every run is driven by a single scalar seed. A mulberry32 generator $s_{k+1} = (s_k + \texttt{0x6D2B79F5}) \bmod 2^{32}$ produces the $(0, 1)$-uniform stream consumed by every stochastic step—dividend realisations, zero-intelligence quotes, the noise term in the utility prior, and the trust-update ordering. Calls to `Math.random` from agent or market code are a contract violation that breaks reproducibility by the (population, seed) pair and are flagged as an invariant in the project style guide. The seed is re-rolled only on Reset, and the replay system reconstructs past states by slicing append-only history arrays on `Market` and `Logger`—no frame is ever recomputed.

IV. Agent Model

This section gives the exact decision rule for each of the five agent types in the simulator. All rules read the top of the book (best bid b_t and best ask a_t), the current fundamental value FV_t, and a seeded pseudo-random draw $r \sim \mathrm{Uniform}(0, 1)$; the formulas below are the literal rules implemented in the source code, with constants exposed to the reader so that any experiment can be reconstructed from this page alone.

A. Fundamentalist (F)

The fundamentalist is a rational mean-reversion anchor inspired by Smith et al. (1988). Its belief is $\beta_t = \mathrm{FV}_t \cdot \big(1 + 0.02\,(r - 0.5) \cdot 2\big)$—a $\pm 2\%$ symmetric noise about the fundamental—and its action rule is

$$\begin{cases} \text{cross the ask with a unit bid at } a_t, & \text{if } a_t < 0.98\,\beta_t \text{ and cash} \geq a_t, \\ \text{cross the bid with a unit ask at } b_t, & \text{if } b_t > 1.02\,\beta_t \text{ and inventory} > 0, \\ \text{post passive with probability } 0.45, & \text{otherwise}, \\ \text{hold}, & \text{otherwise}. \end{cases} \tag{4}$$

Passive quotes bid in $[0.970,\ 1.005] \cdot \beta_t$ and ask in $[0.995,\ 1.030] \cdot \beta_t$, so the fundamentalist never posts a cross on its own initiative. It supplies a narrow two-sided book near the fundamental whenever neither side of the market is aggressive and anchors mispricing back to FV_t through unit crosses when either side drifts by more than 2%.

B. Trend follower (T)

The trend follower estimates a short-horizon slope from the last six non-null trade prices and projects three ticks forward,

$$\text{slope}_t \;=\; \frac{p_t - p_{t-h+1}}{\max(1, h-1)}, \qquad \text{target}_t \;=\; p_t + 3 \cdot \text{slope}_t, \tag{5}$$

where $h = 6$ is the slope window and p_t is the most recent trade price. When $\text{slope}_t > 0$ and it has cash, the trend agent chases up with a unit bid at $p_t \cdot (1 + 0.015 + 0.03\,r)$; when $\text{slope}_t < 0$ and it has inventory, it flees with a unit ask at $p_t \cdot (1 - 0.015 - 0.03\,r)$. Two corner cases protect the agent from stranding. First, if cash has dropped below the most recent trade price but inventory still exceeds the initial endowment, the agent dumps shares into the best bid at the aggressive price. Second, in a flat-slope regime with excess inventory, the agent liquidates with probability $\min(0.85,\ 0.3 + 0.25\,e)$, where e is the excess-inventory rate. The trend rule is the positive-feedback channel (De Long et al., 1990) and is the primary amplifier of bubble formation in the simulator.

C. Random zero-intelligence (R)

The zero-intelligence trader posts uniform bids and asks on fundamentally-scaled support, following Gode and Sunder (1993). Explicitly,

$$\text{bid}_t \sim \text{Uniform}[\,0.7\,\mathrm{FV}_t,\ 1.1\,\mathrm{FV}_t\,], \qquad \text{ask}_t \sim \text{Uniform}[\,0.9\,\mathrm{FV}_t,\ 1.3\,\mathrm{FV}_t\,]. \tag{6}$$

It never reads the order book, never tracks profit, and is pinned in every run so that the book never goes fully empty. One slot of this type is part of the fixed background and is invisible to the population-composition sliders.

D. Experienced (E)

The experienced agent carries the DLM 2005 horizon discount. Let $h = T - t + 1$ be the number of periods remaining. Each experienced agent carries an experience profile $p \in \{\text{naive}, \text{once-played}, \text{veteran}\}$ with a corresponding pair $(\varphi_0^{(p)}, \varphi_1^{(p)})$ of a discount floor and a discount slope, and its belief is

$$\beta_t^{(p)} \;=\; \mathrm{FV}_t \cdot \Big(\varphi_0^{(p)} + \frac{h}{T} \cdot \varphi_1^{(p)}\Big). \tag{7}$$

The three profiles are calibrated as in Table 2.

Table 2: Experienced profiles

Profile	φ_0	φ_1	liquidate $h \leq$	avoid overpay $h \leq$
Naive	1.00	0.05	1	2
Once-played	0.96	0.04	2	3
Veteran	0.92	0.06	3	4

When h falls below the profile's liquidation threshold, the agent dumps any remaining inventory into the best bid; when h falls below the overpay threshold, it refuses any buy above $\beta_t^{(p)}$. Outside the terminal region it behaves like a risk-tight fundamentalist, buying below $\beta_t^{(p)}$ with a discount multiplier (*e.g.* 0.92 for veterans) and selling above $\beta_t^{(p)}$ with a premium multiplier (*e.g.* 1.05). The profile mix is user adjustable, which allows the Experienced population to be dialled between the DLM 2005 subject groups.

E. Utility (U)

The utility agent follows Lopez-Lira (2025). Its prior at every decision tick is

$$V_i^{\text{prior}} \;=\; \text{FV}_t \cdot \big(1 + b_i + \varepsilon_{i,t}\big), \qquad b_i \in \{-0.15,\, 0,\, +0.15\}, \qquad \varepsilon_{i,t} \sim \text{Uniform}[-0.03,\, 0.03]. \tag{8}$$

The persistent bias b_i is the "persistently optimistic/pessimistic/unbiased" archetype and is assigned at sampling time. The per-tick noise ε keeps the agent's quoting out of lockstep with other agents of the same type.

The prior is then blended with incoming messages from peers. If $\mathcal{M}_i$ denotes the set of claimed valuations received during the current period, and $w \in (0, 1)$ is a weight, the posterior is

$$V_i^{\text{post}} \;=\; w \cdot V_i^{\text{prior}} \;+\; (1 - w) \cdot \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \hat{v}_m. \tag{9}$$

Three archetypes determine w: *naive* trusts peers and uses $w = 0.60$; *skeptical* barely moves and uses $w = 0.90$; *adaptive* scales w by per-sender trust and caps the move at 0.50, with pairwise trust updated every trade by

$$\tau_{r \to s} \;\leftarrow\; (1 - \alpha)\, \tau_{r \to s} \;+\; \alpha \cdot \max\!\Big(0,\, 1 - \tfrac{|\hat{v}_s - \text{VWAP}_t|}{\text{VWAP}_t}\Big), \qquad \alpha = 0.30. \tag{10}$$

Receiver r's trust in sender s is reinforced whenever s's claim is close to the period VWAP and attenuated otherwise.

Given the posterior valuation V_i^{post}, the agent scores each available action $a \in \{$hold, cross bid, cross ask, passive bid, pass by its expected utility

$$\text{EU}(a) \;=\; p_{\text{fill}} \cdot U(w_1) \;+\; (1 - p_{\text{fill}}) \cdot U(w_0), \qquad a^\star = \arg\max_a \text{EU}(a), \tag{11}$$

where w_0 and w_1 are the agent's wealth under non-fill and fill respectively, and $p_{\text{fill}} = 0.30$ is a constant passive fill-probability inherited from the original Lopez-Lira design. The risk-preference functional $U \in \{U_L, U_N, U_A\}$ is assigned at sampling time and drawn from

$$U_L(w) = (w/w_0)^2, \qquad U_N(w) = w/w_0, \qquad U_A(w) = \sqrt{w/w_0}. \tag{12}$$

A deceptive variant of the utility rule is available for the messaging channel. When enabled, an asset-poor agent (inventory below its initial endowment) broadcasts a valuation equal to $0.82 \cdot V_i^{\text{true}}$, while an asset-rich agent broadcasts $1.18 \cdot V_i^{\text{true}}$. The lie-gap magnitude $|\hat{v}^{\text{report}} - V^{\text{true}}|$ drives the period-end trust update in (10), which produces the closed-loop feedback described in Lopez-Lira (2025).

Equations (4)–(12) constitute the complete decision model. The market aggregates these orders through a price–time-priority order book, mutates cash and inventory on every trade, credits dividends per (2) at period end, and records every state in append-only history arrays so the replay system can reconstruct past ticks without re-running the agents. No agent in this model consults a language model. The next section adds the first such consultation as the Wang 2026 extension.

V. ChatGPT-Anchored Priors

A. Overview

The Wang 2026 paradigm adds exactly one new moving part to the replicated model of Section IV: a private subjective valuation $\hat{V}_i$, drawn from a ChatGPT call, that replaces the utility agent's deterministic prior on the very first decision tick. Everything downstream of the first tick—belief blending, trust updates, utility maximisation, order matching, dividend payments, and all of the market-quality instruments—is identical to the pure Lopez-Lira run. The anchor is therefore a *minimal* perturbation of the decision model: the smallest change in behaviour that still lets the population's psychology influence its opening bids.

Two design choices make this minimal. First, the call is placed once, in parallel across all utility agents, at run start, and the result is consumed on the first decision tick and then cleared. Second, the call never asks the language model for an *order*; it asks only for a *number*—a valuation per share in experimental cents—which is routed through the same prior slot used by equation (8). The agent's action is still chosen by (11), using a wealth argument that the language model never sees. This construction isolates the psychological channel from the trading channel; it also keeps the downstream replay deterministic, because after the first tick no agent in Wang 2026 depends on external randomness of any kind.

B. The anchor call

Let A_U be the set of utility agents in the current population. For each $i \in A_U$ the simulator constructs a short prompt π_i that encodes three facts: the paper constants $(T, \mathbb{E}[d])$, the agent's risk preference $U_i \in \{U_L, U_N, U_A\}$ and bias mode $b_i \in \{\text{over}, \text{none}, \text{under}\}$, and the fact that the market is shared with DLM 2005 experienced counterparties. The system prompt is fixed and instructs the model to return a single number with no explanation, currency symbol, or prose. Concretely, the user prompt template reads

> Market: T periods. Each share pays 0 or 20 cents with equal probability per period, expected dividend $\mathbb{E}[d]$. Risk-neutral fundamental value at run start is FV_0 cents (declines by $\mathbb{E}[d]$ per period). You, agent i, are *risk* and *bias*. You share the market with Experienced agents (Dufwenberg, Lindqvist and Moore 2005) who know the asset decays to zero and refuse to overpay in late periods. What is your private subjective valuation per share right now, in cents?

where risk $\in \{\text{risk-loving}, \text{risk-neutral}, \text{risk-averse}\}$ and bias $\in \{\text{persistently optimistic}, \text{unbiased}, \text{persistently pessimistic}\}$. All calls are fired in parallel as a single `Promise.all` batch, so the network latency is one round trip rather than $|A_U|$ round trips.

The raw reply is parsed for the first signed decimal, parsed as a float, and clamped to

$$\hat{V}_i \in [\,0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0\,] \;=\; [25,\ 175], \tag{13}$$

so that any out-of-range reply is bounded and cannot destabilise the downstream engine. A missing reply, a parse failure, or an HTTP error silently drops the anchor for that agent; its prior then defaults to equation (8) on the first tick, and the run is otherwise unaffected. This deterministic fallback preserves the invariant that Wang 2026 produces a simulation with or without network connectivity—a requirement for the design to be reproducible and shareable.

C. Consumption on the first decision tick

On every utility agent's first call to `updateBelief` after the simulation starts, the "psych anchor" slot is consulted in a *consume-and-clear* step:

$$V_i^{\text{prior}} \;=\; \begin{cases} \hat{V}_i \cdot (1+\varepsilon_i) & \text{if } \hat{V}_i \neq \emptyset, \\ \mathrm{FV}_t \cdot (1+b_i+\varepsilon_i) & \text{otherwise,} \end{cases} \qquad \text{then set } \hat{V}_i \leftarrow \emptyset. \tag{14}$$

The first branch overrides the deterministic prior with the language-model anchor and keeps the Lopez-Lira noise $\varepsilon_i \sim \text{Uniform}[-0.03, 0.03]$ so that two utility agents assigned identical anchors still draw slightly different first quotes. The second branch is the pure Lopez-Lira prior, used on every tick after the first and on every tick when no anchor is available. The "clear" step ensures that a second call to `updateBelief` on the same tick, or the next decision tick in the same run, goes through the standard path, so that the language model's effect is felt at run start and then decays through the belief-blend equation (9) as normal messaging proceeds.

D. Research question and instrumentation

Let $i^\star = \arg\max_{i \in A_U} \hat{V}_i$ be the utility agent with the highest anchored valuation, and define the top holder at tick t as

$$j^\star(t) \;=\; \arg\max_{i \in A_U} \text{inventory}_i(t). \tag{15}$$

The research question is simply whether $j^\star(t) \to i^\star$ by the end of the run. The simulator reports, every tick, a boolean *match* indicator $\mathbb{1}\{j^\star(t) = i^\star\}$ and a *normalised gap*

$$g(t) \;=\; \frac{\hat{V}_{i^\star} - \hat{V}_{j^\star(t)}}{\hat{V}_{i^\star}} \;\in\; [0,1], \tag{16}$$

together with the identities of the two agents and their valuations. The gap collapses to zero when the asset has reached the psychologically-maximal holder; it stays near one when the asset has migrated to an agent whose anchor is much lower than the maximum. Figure placeholders in the `figures/` directory will host time-series plots of $g(t)$ and the match indicator for each of the three paradigms.

The match indicator is related, but not identical, to the allocative efficiency statistic

$$\mathrm{AE} \;=\; \frac{\sum_i \hat{V}_i \cdot \mathrm{inventory}_i(T)}{\max_i \hat{V}_i \cdot Q}, \tag{17}$$

in which the numerator is the realised cross-agent utility weighted by each agent's anchored valuation and the denominator is the best-possible allocation in which every share is held by the maximum-valuation agent. When $\mathrm{AE} = 1$ the asset has migrated to $i^\star$ and the match indicator is true; but AE can be close to one even when the match indicator is false, for example when the top holder is different from $i^\star$ but has a nearly-maximal anchor. Reporting both statistics together is therefore informative: the match indicator captures the identity of the eventual owner, while AE captures the continuous loss relative to the psychological optimum.

E. Determinism, cost, and caching

A single Wang 2026 run consumes $|A_U|$ chat completions as one parallel batch and runs closed-form thereafter. With $|A_U| \leq 6$ in the default preset, network cost is a handful of round trips on the first frame and zero afterwards; no caching or retry layer is needed because a missing anchor silently degrades to the deterministic prior. The seeded PRNG makes identical anchor vectors reproduce identical trajectories, so within-model replay is trivial and across-model comparisons reduce to toggling the paradigm button.

VI. Preliminary Results

Because the simulator is designed for interactive replication rather than a single canonical experiment, this section reports preliminary findings from three paradigm presets rather than a calibrated empirical test. Each paradigm holds the market structure of Section III fixed and varies only the population composition. Unless otherwise noted, $N = 10$ agents share one market for $T = 10$ periods, the seeded PRNG is rolled fresh at the start of every run, and the Wang 2026 anchor is drawn from the default model `gpt-5.4` with the paper's system and user prompts verbatim.

A. Baseline DLM 2005: experience kills the bubble

Under the Experienced preset ($3\,E + 2\,F + 1\,T$), transaction prices track the fundamental staircase within a few experimental cents from period 1; volume concentrates in the first three periods and decays; amplitude stays below 0.15, ND below 0.10, and Haessel's R^2 is typically above 0.90. The naive experienced profile ($\varphi_0 = 1.00, \varphi_1 = 0.05$) yields a weak bubble-killing channel and modest overshooting in periods 3–5; the veteran profile ($\varphi_0 = 0.92, \varphi_1 = 0.06$, liquidate at $h \leq 3$) sharpens convergence to below one experimental cent. The qualitative DLM 2005 finding—that experience, rather than instructions or incentives, kills the bubble—is reproduced.

B. Lopez-Lira 2025: the strategy cube shapes the bubble

Under the Lopez-Lira preset (six utility agents, no experienced), the price dynamic depends sensitively on the population's *strategy cube*: bias mode $\times$ belief mode $\times$ risk preference. When $\alpha_L > 0.6$, the convex $U_L(w)$ drives aggressive ask-crossing in early periods with amplitude $A > 0.40$, positive residuals through periods 1–7, and a terminal collapse—the canonical SSW bubble-and-crash shape with Haessel R^2 near zero. When $\alpha_A > 0.6$, concave $U_A(w)$ pulls residuals negative and raises ND into the 0.15–0.25 range, reflecting under-pricing rather than bubble formation. Toggling the deceptive messaging mode produces the largest per-

turbation: deceptive broadcasters drag the trust EMA down once the period-end closeness update of (10) runs, and the mean-lie-magnitude statistic is tightly correlated with ND, consistent with the closed-loop mechanism in Lopez-Lira (2025).

C. Wang 2026: does the asset migrate to the highest anchor?

The novel regime is the Wang 2026 merged market. With $\lceil (N-4)/2 \rceil = 3$ experienced and 3 utility agents and a live ChatGPT anchor vector $(\hat{V}_i)_{i \in A_U}$, three preliminary observations are clear from a handful of runs.

Anchor dispersion.. The returned anchors are heterogeneous across risk and bias archetypes in the expected direction. Risk-loving, persistently-optimistic agents receive anchors near the upper clamp $1.75\,\mathrm{FV}_0 = 175$; risk-averse, persistently-pessimistic agents receive anchors near the lower clamp $0.25\,\mathrm{FV}_0 = 25$; risk-neutral, unbiased agents cluster near $\mathrm{FV}_0 = 100$. The simulator reports the full anchor vector in its subjective-valuation panel, so the reader can verify that the model's psychology maps into $\hat{V}$ in the direction the prompt implies.

First-tick quoting.. Because the anchor is consumed on the first decision tick, the earliest orders in a Wang 2026 run are visibly stratified by anchor. Risk-loving agents with $\hat{V} \approx 175$ post aggressive bids near $\mathrm{FV}_1 = 100$ and cross the ask when possible; risk-averse agents with $\hat{V} \approx 25$ post asks well below FV_1. The effect lasts roughly one period before the Lopez-Lira belief-blending channel pulls everyone back toward FV_t, but it produces a clear opening asymmetry absent from a pure Lopez-Lira run with the same seed.

The match indicator.. Across runs, the boolean match indicator $\mathbb{1}\{j^\star(t) = i^\star\}$—whether the agent currently holding the most shares is the agent with the highest anchor—is not monotone in t. It typically starts near the seed endowment distribution (which is independent of $\hat{V}$), rises during periods 1–3 as the aggressive high-anchor agents accumulate inventory, and then drifts as the experienced agents liquidate in the terminal region. In runs where the highest-anchor agent is also the experienced bubble's counterparty, the indicator ends at one; in runs where the experienced block liquidates into an agent of middling anchor, the indicator ends at zero and the corresponding normalised gap $g(T)$ is typically in the 0.10–0.30 range. The allocative efficiency statistic AE moves in the same direction but stays strictly positive in every run because the denominator in (17) is $\max_i \hat{V}_i \cdot Q$ and the numerator always contains a nontrivial contribution from the top holder.

D. Market-quality statistics at a glance

Table 3 summarises the canonical bubble statistics under the three paradigms in qualitative—not yet calibrated—form.

Table 3: Market-quality statistics, qualitative ranges

	DLM 2005	Lopez-Lira 2025	Wang 2026
Haessel R^2	> 0.90	0.30–0.80	0.40–0.85
Normalised dev. ND	< 0.10	0.10–0.30	0.10–0.25
Amplitude A	< 0.15	0.20–0.50	0.20–0.45
Turnover TO	0.4–0.8	0.6–1.2	0.5–1.0
Allocative eff. AE	≈ 1	0.6–0.9	0.7–0.95
Match $\mathbb{1}\{j^\star(T) = i^\star\}$	n/a	n/a	50–70% runs

Wang 2026 sits between its two parents: the experienced block suppresses amplitude and improves R^2, while the utility block introduces enough psychology to produce nontrivial variation in the match indicator. Whether the match indicator exceeds a naive baseline in a properly replicated experiment—and whether the anchor adds predictive power beyond the deterministic prior—is the question the testbed is built to answer.

VII. Conclusion

This paper documents a browser-based replication of the experimental asset market of Dufwenberg et al. (2005), extended with the expected-utility messaging population of Lopez-Lira (2025), and introduces a new

paradigm in which a single ChatGPT call seeds each utility agent's private subjective valuation at run start. The extension is deliberately minimal: one number per agent, clamped to $[0.25\,\mathrm{FV}_0,\ 1.75\,\mathrm{FV}_0]$, written to the `psychAnchor` slot and consumed on the first decision tick of the standard Lopez-Lira prior. Everything downstream of the first tick—the belief blend, the trust EMA, the expected-utility action choice, the order matching, the dividend crediting, and the market-quality instruments—is unchanged from the replicated baseline, and the simulator's seeded mulberry32 generator ensures that every run can be replayed identically with or without network access.

The contribution is threefold. First, the testbed constitutes a clean vanilla-JavaScript implementation of two otherwise unrelated experimental designs, specified in enough detail (Sections III–IV) that a reader can reconstruct the model from the paper alone. Second, the Wang 2026 extension introduces a narrowly-scoped use of a language model as a *psychological prior elicitor* rather than as a trader: the model is never asked to choose an action, and its output is routed through the same prior slot used by the deterministic fallback, so that any behavioural effect is attributable to the starting valuation rather than to an expansion of the action space. Third, the research question—does the asset end up in the hands of the agent with the highest anchored valuation—is instrumented with a per-tick match indicator and a normalised gap, and reported alongside the classical bubble statistics (Haessel R^2, normalised deviation, amplitude, turnover, allocative efficiency). The instrumentation is designed so that any replication of the Wang 2026 paradigm can report the same panel without additional code.

Three caveats shape the preliminary findings. First, the ranges in Table 3 are illustrative, not calibrated: a full empirical test requires many replications per cell, a pooled treatment of the network variance of the anchor, and a formal baseline against a deterministic prior with matched mean and variance. Second, the anchor is consumed exactly once, which bounds the downstream signal at whatever the first-tick prior contributes to the five-action expected-utility score; refreshing $\hat{V}_i$ each period would be a natural extension but is in tension with the reproducibility invariant. Third, generalising beyond the DLM 2005 dividend distribution and horizon would require re-deriving the market-quality instruments and re-calibrating the experienced profiles of Table 2.

The testbed is free of build steps and of third-party dependencies, and ships with the architecture diagram, glossary, and slide deck used in this paper. Its "Experiment settings" panel exposes every numerical constant that shapes a run—the population mix, the risk-preference share, the passive fill probability, the trust learning rate, the bias amplitude, and the valuation noise—and the seeded PRNG makes every edit to a constant observable as a difference against a matched baseline. The Wang 2026 paradigm is the first extension of this infrastructure that asks a question about language models, and it opens several follow-ups: whether the anchor's effect on allocative efficiency scales with the cross-agent variance of $\hat{V}$; whether multiple language models produce systematically different anchor vectors for the same prompt; and whether adding deceptive messaging amplifies or dampens the first-tick stratification that the anchor produces.

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